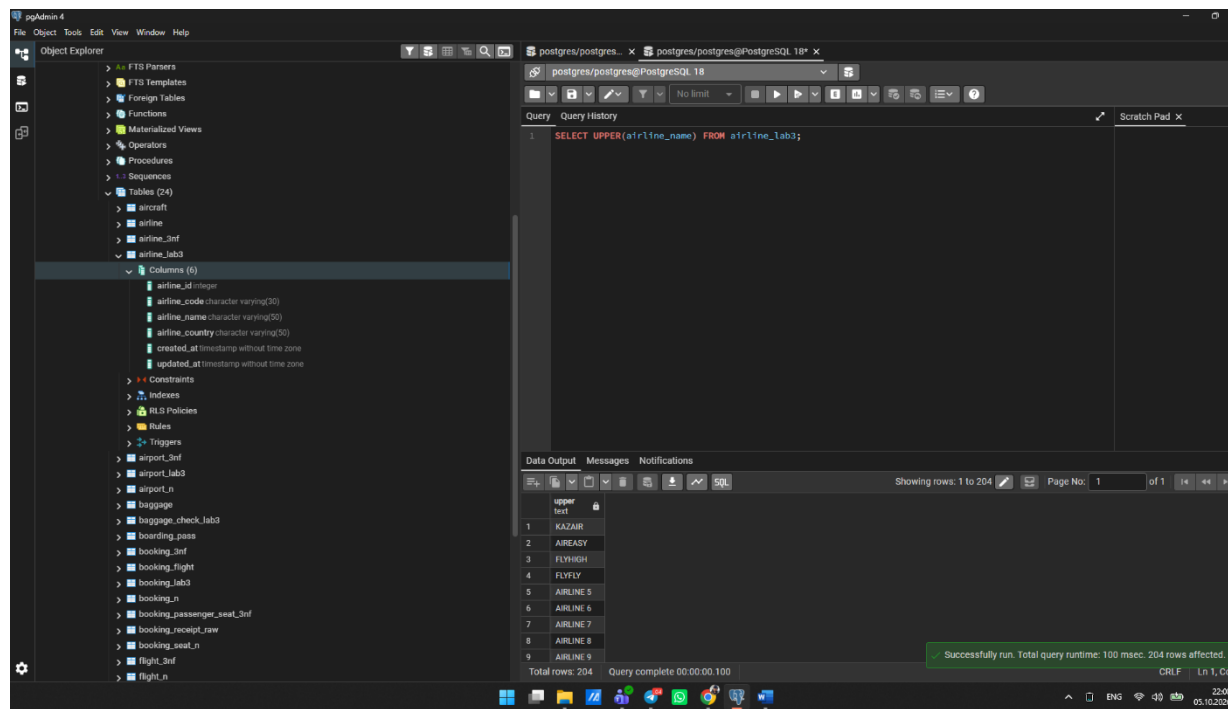


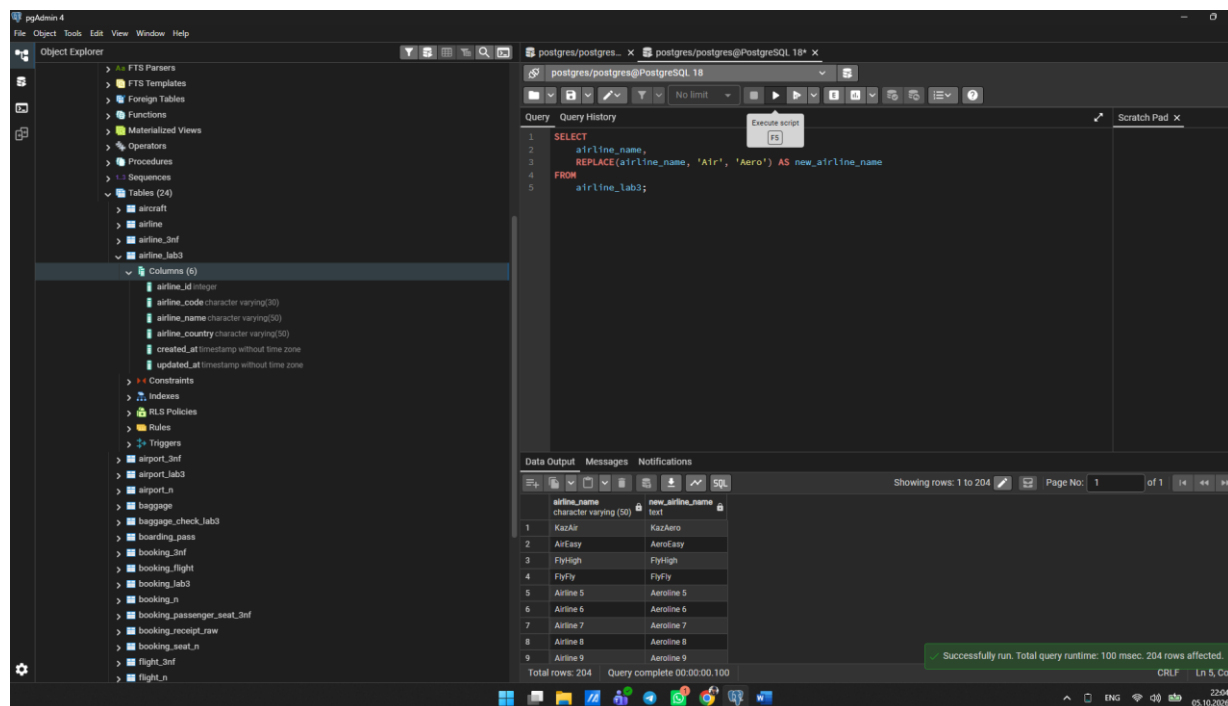
Laboratory work 4

Tasks:

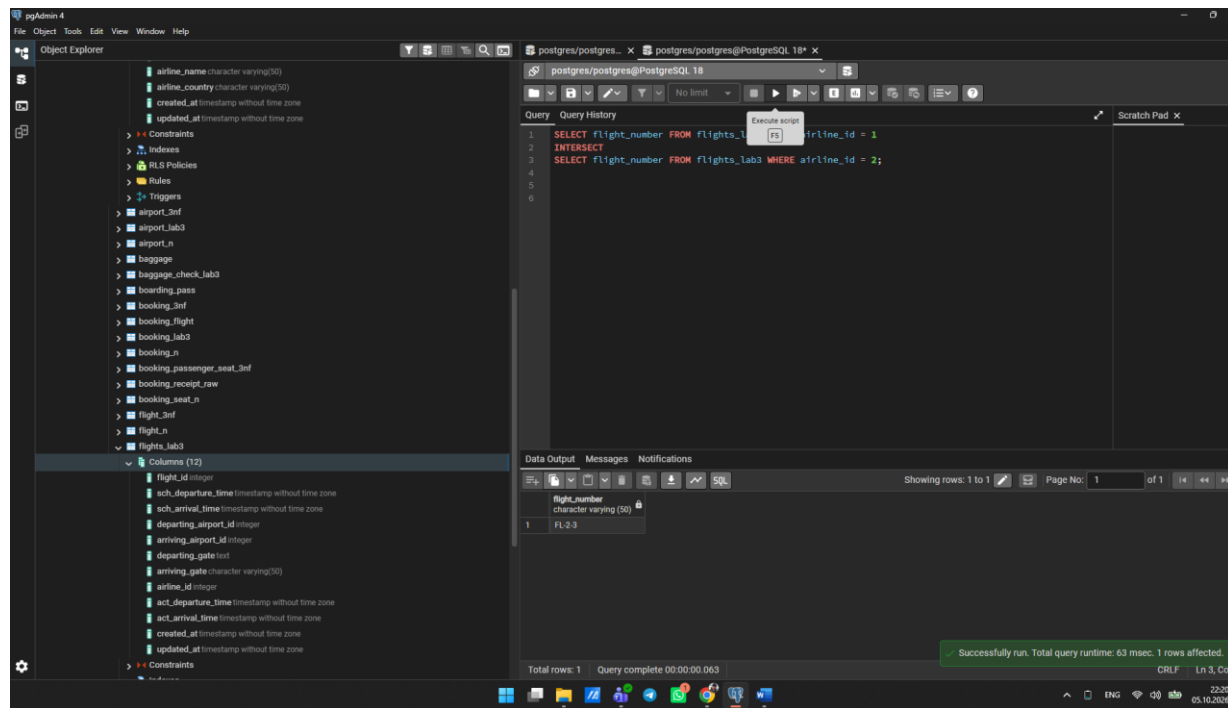
1. Retrieve all airline names in uppercase.



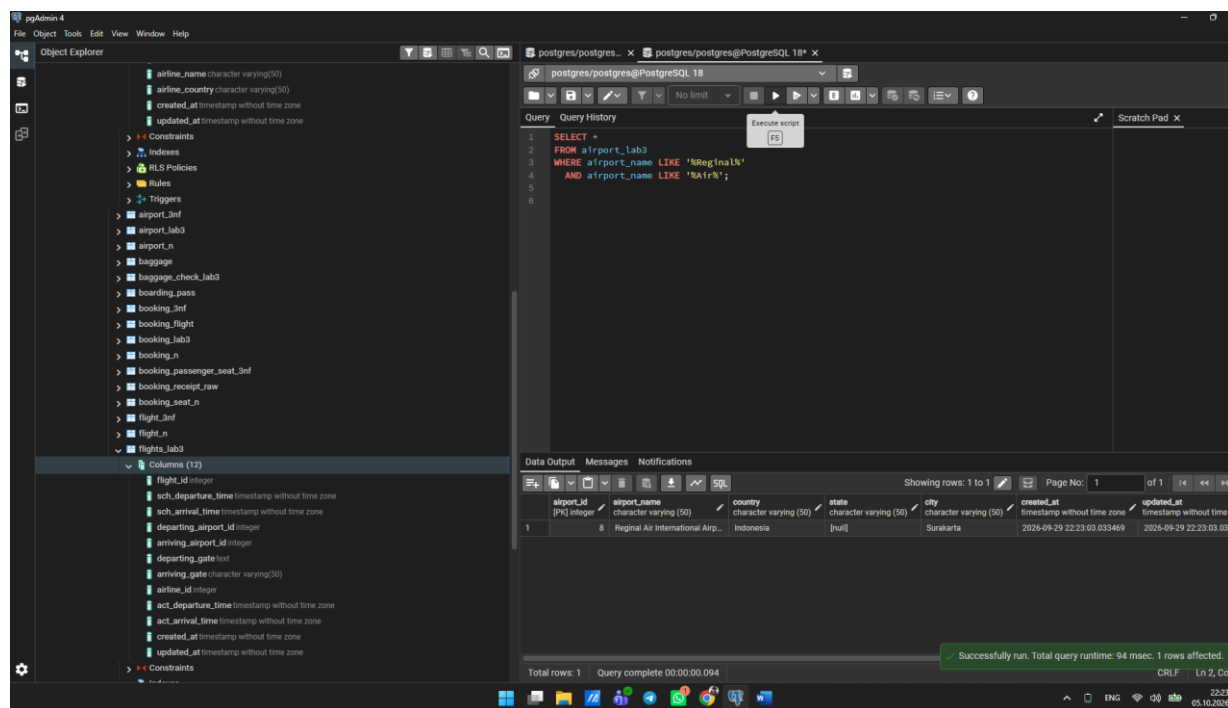
2. Replace any occurrence of the word "Air" in airline names with "Aero".



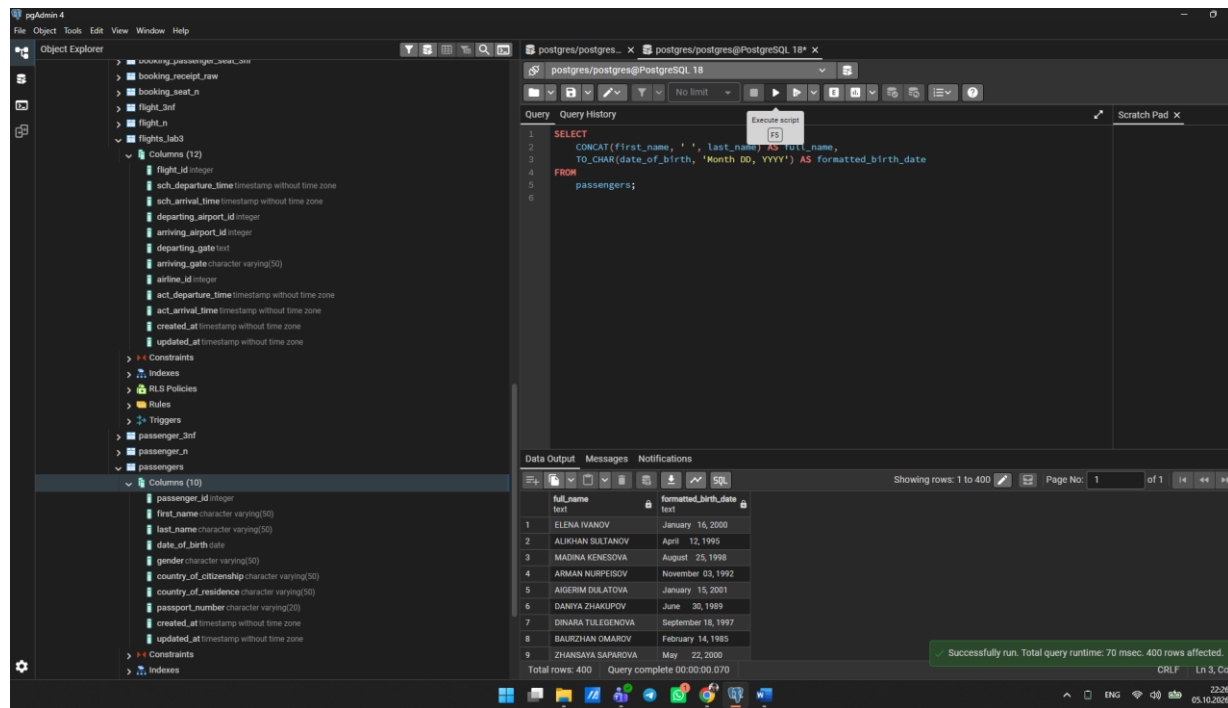
3. Find all flight numbers that coordinates with both airline 1 and airline 2.



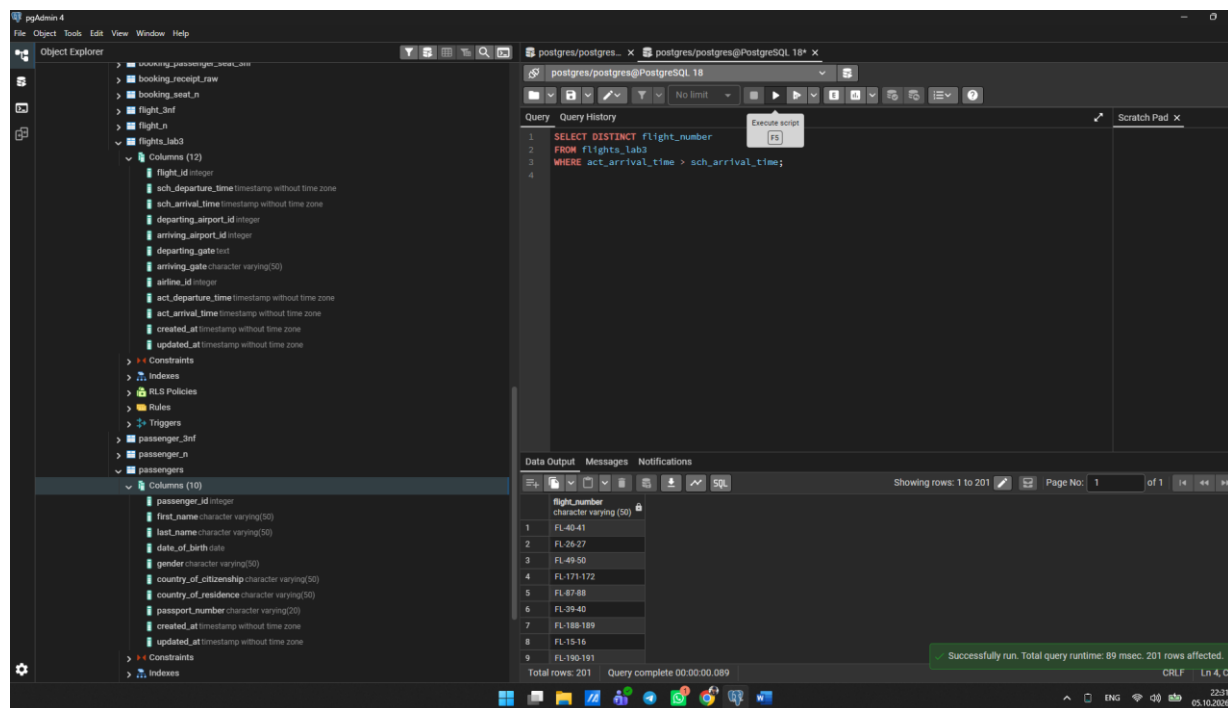
- Retrieve airports that contain the word "Reginal" and "Air" in their names.



- Retrieve passenger names and format their birth dates as 'Month DD, YYYY'.



- Find flight numbers that have been delayed based on the actual arrival time.



- Find flights that arrived late according to their actual arrival time compared to the scheduled arrival time.

The screenshot shows the pgAdmin 4 interface. On the left, the 'Object Explorer' pane displays the database structure, including tables like 'booking_receipt_raw', 'booking_seat_n', 'flight_3nf', 'flight_n', and 'flights_lab3'. The 'flights_lab3' table is selected, showing its columns: flight_id, sch_departure_time, sch_arrival_time, departing_airport_id, arriving_airport_id, departing_gate, arriving_gate, airline_id, act_departure_time, act_arrival_time, created_at, and updated_at.

The main pane shows a SQL query executed in the 'Query History' tab:

```

1 SELECT
2   flight_id,
3   departing_airport_id,
4   arriving_airport_id,
5   sch_arrival_time,
6   act_arrival_time,
7   (act_arrival_time - sch_arrival_time) AS lateness_duration
8 FROM
9   flights_lab3
10 WHERE
11   act_arrival_time > sch_arrival_time;
12

```

The 'Data Output' pane shows the results of the query, displaying columns: flight_id, departing_airport_id, arriving_airport_id, sch_arrival_time, act_arrival_time, and lateness_duration. The results show 201 rows of data.

At the bottom, a status bar indicates: 'Total rows: 201 Query complete 00:00:00.101 Successfully run. Total query runtime: 101 msec. 201 rows affected.'

8. Show airlines from any of: ('France','Portugal','Poland') created between '2023-11-01' and '2024-03-31'.

The screenshot shows the pgAdmin 4 interface. On the left, the 'Object Explorer' pane displays the database structure, including tables like 'booking_receipt_raw', 'booking_seat_n', 'flight_3nf', 'flight_n', and 'flights_lab3'. The 'flights_lab3' table is selected, showing its columns: flight_id, sch_departure_time, sch_arrival_time, departing_airport_id, arriving_airport_id, departing_gate, arriving_gate, airline_id, act_departure_time, act_arrival_time, created_at, and updated_at.

The main pane shows a SQL query executed in the 'Query History' tab:

```

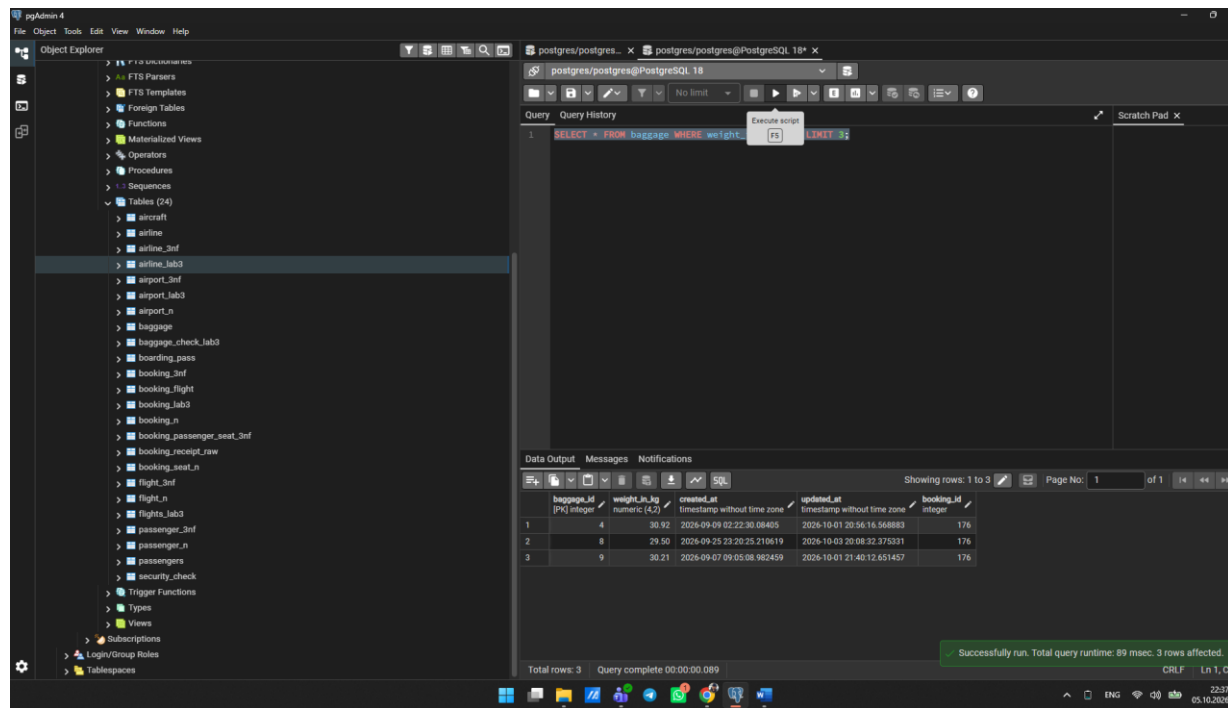
1 SELECT
2   airline_name,
3   airline_country,
4   created_at
5 FROM
6   airline_lab3
7 WHERE
8   airline_country IN ('France', 'Portugal', 'Poland')
9   AND created_at BETWEEN '2023-11-01 00:00:00' AND '2024-03-31 23:59:59';
10

```

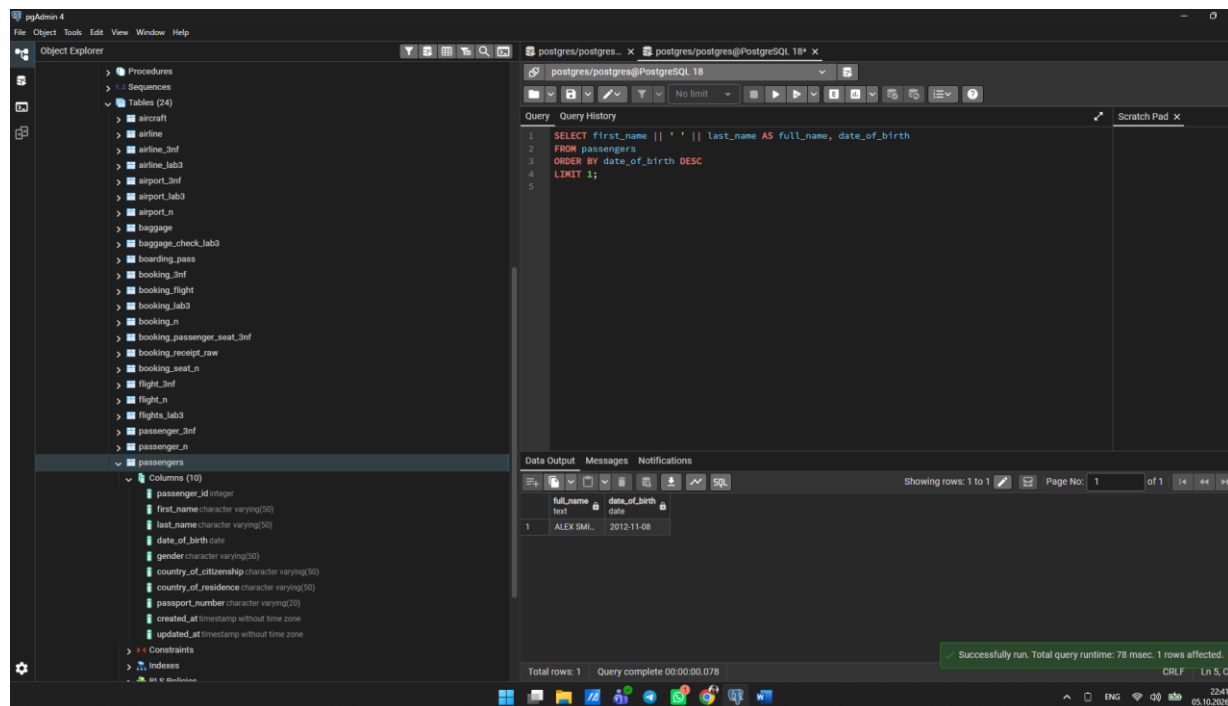
The 'Data Output' pane shows the results of the query, displaying columns: airline_name, airline_country, and created_at. The results show 0 rows of data.

At the bottom, a status bar indicates: 'Total rows: 0 Query complete 00:00:00.072 Successfully run. Total query runtime: 72 msec. 0 rows affected.'

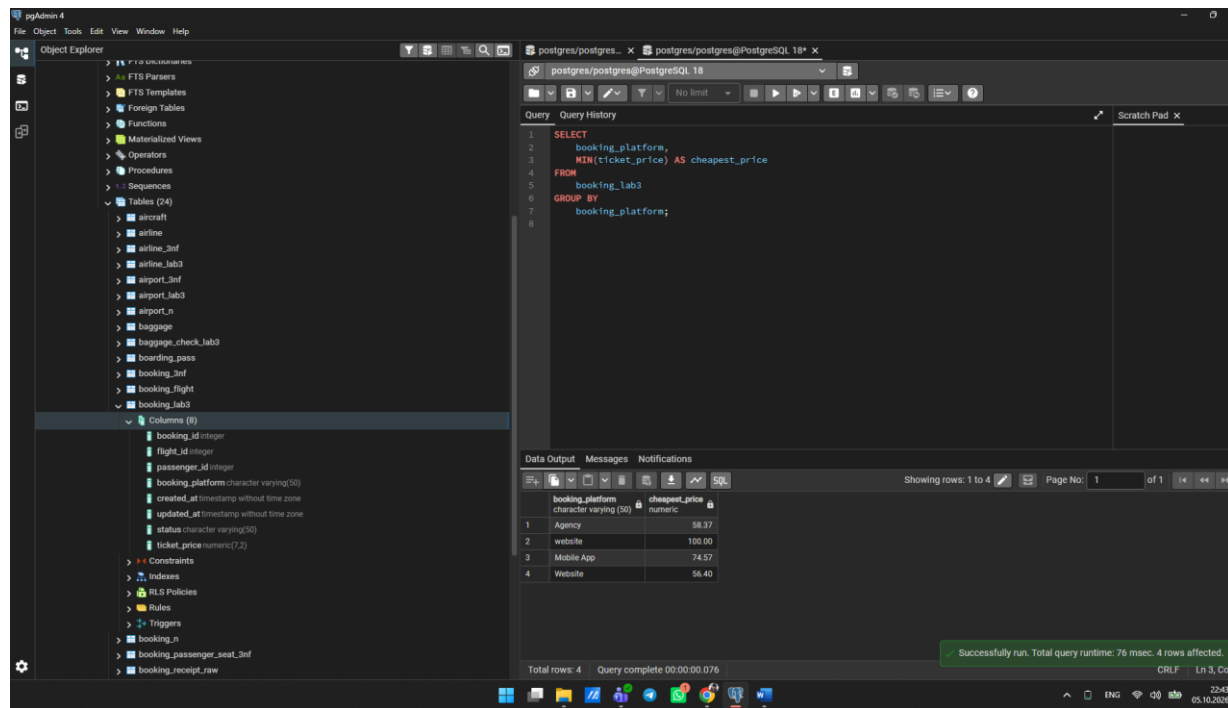
9. Find top3 overweight baggage with more than 25kg.



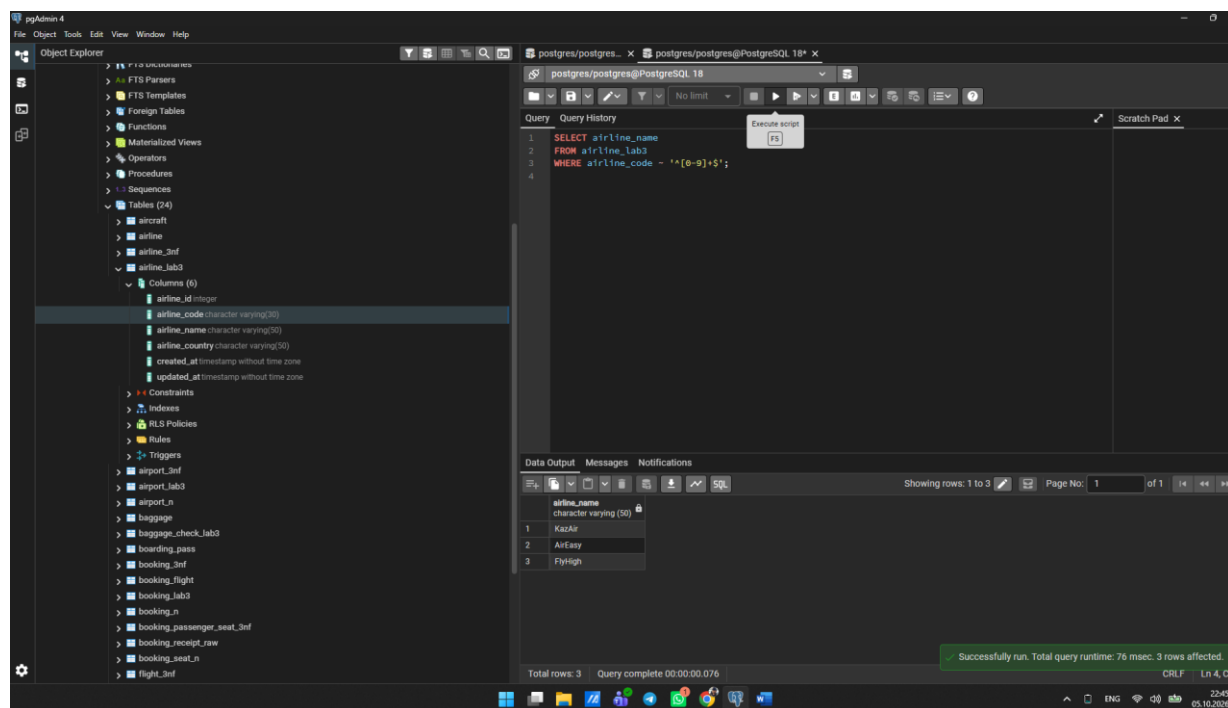
10. Find the youngest passengers' full name.



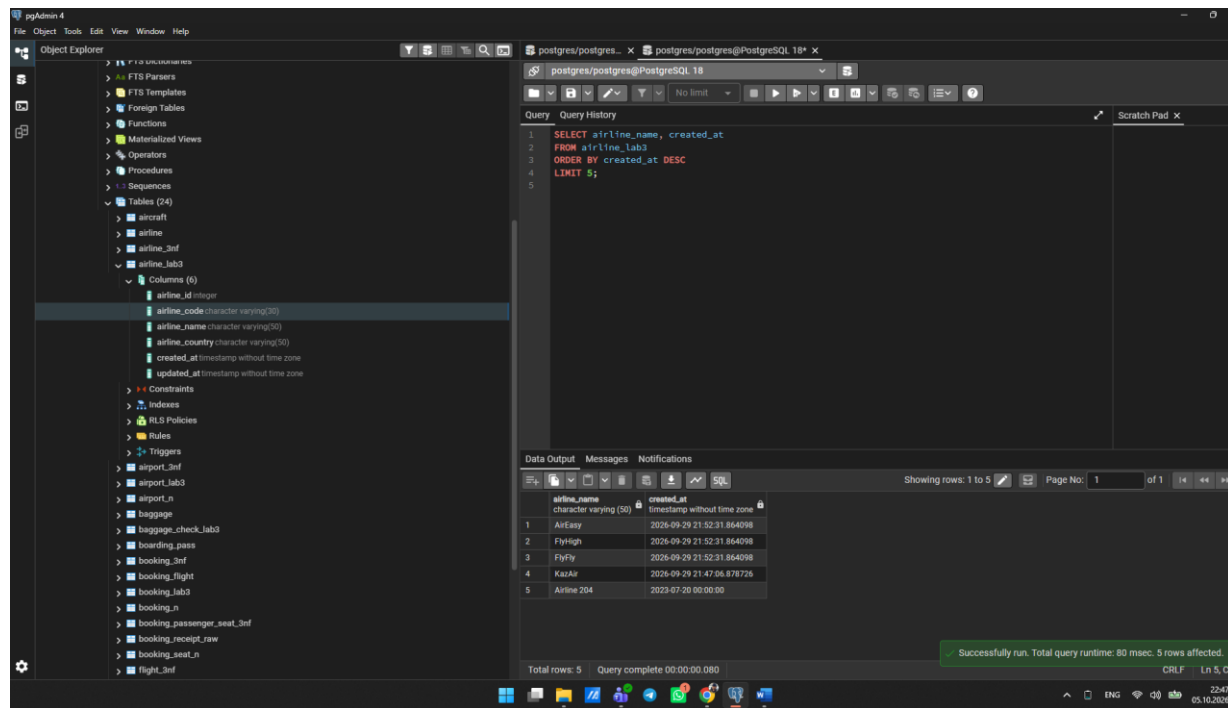
11. Find the cheapest booking price on each booking platform.



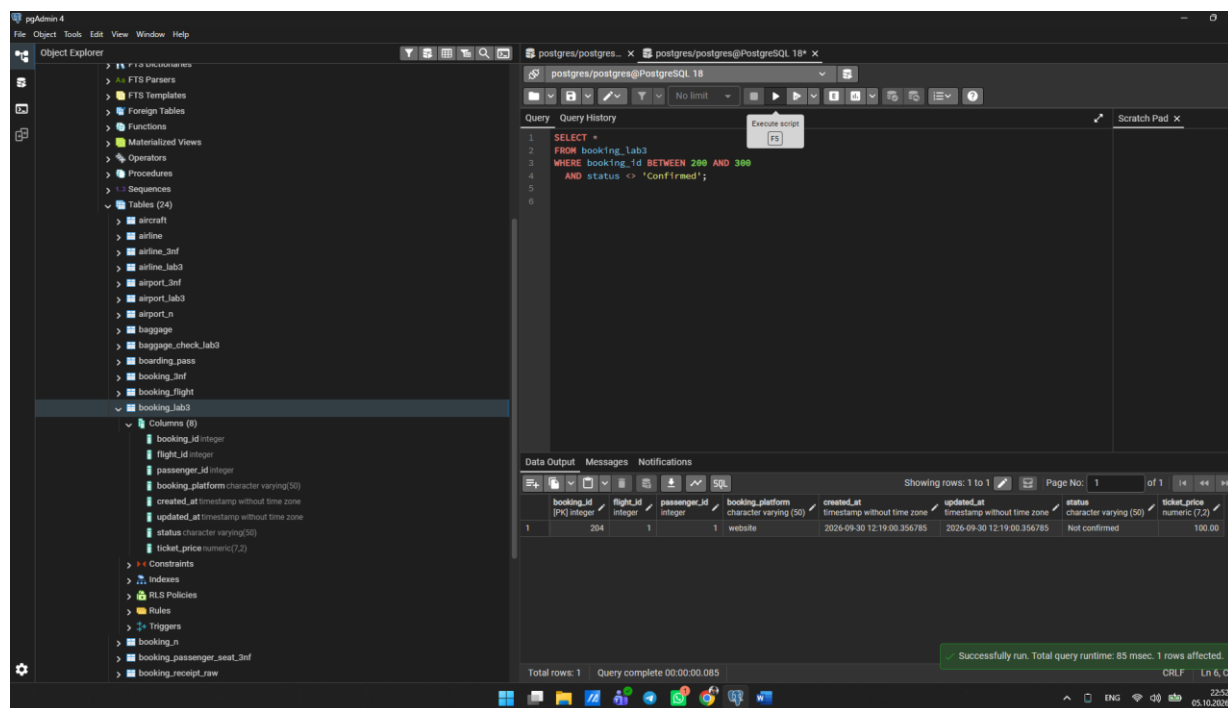
12. Return airlines whose airline_code contains a digit.



13. List the top5 most recently created airlines.



14. Return all rows where booking_id is between 200 and 300 inclusive and check_result <> 'Checked'.



15. Baggage checks where update_at is in the same month as created_at but occurs earlier than created_at.

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- FTS Parameters
- FTS Templates
- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (24)
 - aircraft
 - airline
 - airline_3nf
 - airline_lab3
 - airport_3nf
 - airport_lab3
 - airport_n
 - baggage
 - baggage_check_lab3
 - boarding_pass
 - booking_3nf
 - booking_flight
 - booking_lab3
 - booking_n
 - booking_passenger_seat_3nf
 - booking_receipt_raw
 - booking_seat_n
 - flight_3nf
 - flight_n
 - flight_lab3
 - passenger_3nf
 - passenger_n
 - passengers
 - security_check
- Trigger Functions
- Types
- Views
- Subscriptions
- Login/Group Roles
- Tablespaces

postgres/postgres... X postgres/postgres@PostgreSQL 18+ X

postgres/postgres@PostgreSQL 18

Query Query History

```
1 SELECT *
2 FROM baggage_check_lab3
3 WHERE DATE_TRUNC('month', updated_at) = DATE_TRUNC('month', created_at)
4 AND updated_at < created_at;
5
```

Data Output Messages Notifications

baggage_check_id	check_result	created_at	updated_at	booking_id	passenger_id
[PK] integer	character varying (50)	timestamp without time zone	timestamp without time zone	integer	integer

Total rows: 0 Query complete 00:00:00.078

Successfully run. Total query runtime: 78 msec. 0 rows affected.

CRLF Ln 2, Co

22:54 05.10.2024